NAMMA-METRO SAHAYA

SOFTWARE REQUIREMENTS DOCUMENT (SRD)

Android App Development using GenAI | Metro Infrastructure

MindMatrix VTU Internship Programme | Project Title: 73

First-Timer's Metro Guide · Visual Directions · Offline-Ready

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Document Contents

1. Introduction and Purpose

1.1 Purpose of This Document

This Software Requirements Document (SRD) defines the complete, precise, and testable requirements for the Namma-Metro Sahaya Android application. It expands the MindMatrix PRD into:

- Atomic Functional Requirements (FR): Each prefixed FR-NMS-XX, written as 'The system shall'
- Measurable Non-Functional Requirements (NFR): Quality constraints with specific test thresholds from PRD Section 6
- Use Cases: 12-field specifications for each major user interaction
- Requirements Traceability: Every FR linked to a Use Case, Design Component, and Test Case ID

This document becomes the authoritative reference for development, testing, and stakeholder acceptance. Any change after mentor sign-off requires a formal Change Request per the Project Charter (PC-NMS-001).

1.2 Scope

In Scope

- F-01: Simple route query — user enters source and destination, app computes Metro path
- F-02: Step-by-step Visual Guide with platform photos and directional instructions
- F-03: Exit Finder — identifies the correct exit gate for a given destination (e.g., KSRTC Bus Stand)
- F-04: Fare & Time display — ticket price and estimated travel time for the journey
- F-05: Interchange guidance — clear Purple/Green line switch instructions at Majestic
- F-06: Token machine walkthrough — step-by-step card tapping and token purchase guide
- F-07: Offline mode — station exit images and route data cached locally for tunnel use
- F-08: Kannada language UI — large-font Kannada text for all primary screens

Out of Scope

- OOS-01: Real-time train arrival tracking — no official BMRCL API available; Phase 2 roadmap
- OOS-02: Ticket purchase or payment integration — not part of MVP
- OOS-03: iOS / web platform — Android only for internship MVP
- OOS-04: Bus or auto-rickshaw routing beyond Metro network
- OOS-05: Live crowd density or coach occupancy data

1.3 Intended Audience

Reader	Purpose for Reading This Document
Development Team (UI Dev, DB Handler, Feature Integrator)	Understand exact requirements before coding. Every FR is a development target. Every UC is a coding specification.
Project Sponsor / Mentor	Review and accept requirements. Sign off Section 11. Basis for milestone demo evaluation.
MindMatrix Programme Evaluator	Verify that Section 16 evaluation criteria are met. The RTM (Section 10) links evaluation criteria to implemented features.
Tester (Week 5)	Derive test cases from FRs, Exception Flows, and NFR acceptance tests in Sections 4, 5, and 6.

2. Stakeholders and User Personas

2.1 Stakeholder Map

Stakeholder	Role	Power/Interest	Key Influence
Mentor (Project Sponsor)	Approves SRD. Accepts deliverables at Week 6.	HIGH / HIGH	Defines acceptance criteria. Final authority on Change Requests.
DB Handler	Implements Room DB / SQLite entities per Section 8.	TEAM / TECH	Section 8 Data Requirements authored and owned by this role.
UI Developer	Implements all screens per Section 9.	TEAM / TECH	Section 9 UI/UX Requirements and Use Case triggers/screens.
Feature Integrator	Implements GenAI, pathfinding, offline caching.	TEAM / TECH	Section 4 FRs for route logic, Visual Guide, and offline features.
End User — Meena Devi (Rural First-Timer)	Rural passenger using Metro for the first time.	LOW / HIGH	Meena's confusion drives all Visual Guide and Exit Finder FRs.
End User — BMRCL Admin	Updates station exit and interchange data.	LOW / HIGH	Admin data entry FRs and dashboard UX.

2.2 User Personas

Persona 1 — Meena Devi (Primary First-Time Metro User)

Name / Age	Meena Devi | Age 34
Occupation	Vegetable vendor from Tumkur, travelling to NIMHANS for a medical appointment
Goal	Board the correct Metro line, know where to exit, and reach NIMHANS without getting lost
Pain Points	Cannot read English signage; confused by token machines; afraid of taking the wrong exit or missing the interchange at Majestic
Expectation	App shows photos of her platform, step-by-step in Kannada, and tells her exactly which gate leads to the hospital
Scenario	Meena enters 'Majestic' as source and 'Nimhans' as destination. App displays: take Green Line to Jayanagar, exit Gate 2, walk 200m. Each step has a platform photo. Works offline inside the tunnel.
Derived Requirements	FR-NMS-01 (route query) | FR-NMS-03 (visual guide) | FR-NMS-04 (exit finder) | FR-NMS-07 (offline) | FR-NMS-08 (Kannada UI)

Persona 2 — BMRCL Station Data Admin

Name / Age	Rajesh Kumar | Age 42
Occupation	Metro station operations supervisor
Goal	Keep station exit data, platform photos, and interchange info accurate and up to date
Pain Points	No digital tool to update exit gate associations; gate numbers change seasonally
Scenario	Rajesh updates Gate 3 of Jayanagar station to point to NIMHANS instead of Gate 2 after construction changes. All users immediately see the correct gate.
Derived Requirements	FR-NMS-09 (admin station edit) | FR-NMS-10 (exit gate management)

3. System Context and Architecture Overview

3.1 System Architecture

Layer	Components	Role	Reference
View (UI Layer)	All Jetpack Compose Composable functions (Stepper screens, Route Input, Exit Finder)	Renders state. No business logic. No direct DB calls.	View Layer — Compose screens
ViewModel Layer	RouteViewModel, GuideViewModel, ExitViewModel, FareViewModel	Exposes StateFlow to UI. Contains pathfinding and ETA logic.	ViewModel layer
Repository Layer	RouteRepository, StationRepository, ExitRepository	Abstracts all data operations. Mediates between ViewModel and Room DB.	Repository classes handle CRUD
Data Layer (Room DB)	Entity classes + DAO + AppDatabase	Local storage for stations, exits, routes, photos. Schema in Section 8.	Local DB: Room
GenAI Layer	Gemini API / on-device model for natural language query parsing	Parses free-text destination into structured station lookup.	Cloud API: Gemini

3.2 Architecture Constraint

CRITICAL: No Room DAO or Gemini API method shall be called directly from any Composable function. All data access must route through ViewModel → Repository → Room/API. This constraint is evaluated in Section 16 Code Quality (20%) and tested in NFR-MAINT-01.

4. Functional Requirements (FR)

4.1 FR Format and Conventions

Every Functional Requirement follows this format:

FR-NMS-[Number]: The system shall [verb] [object] [conditions] [constraint / threshold].

- FR-NMS-XX: Namma-Metro Sahaya functional requirement
- Priority codes: M = Must Have | S = Should Have | C = Could Have (Innovation)
- Every FR is: Atomic (one behaviour per FR) | Testable | Unambiguous (no ‘fast’, ‘easy’, ‘good’)

4.2 FR Group 1 — Route Query and Path Calculation (PRD Section 5.1)

FR ID	Requirement — The system shall...	Pri	UC Ref	Test IDs
FR-NMS-01	...allow a user to enter a source station and destination station (free-text or dropdown). On submission, the system shall compute the shortest Metro path using a graph-based pathfinding algorithm (Nodes = Stations, Edges = track segments) and display the step-by-step journey within 2 seconds.	M	UC-NMS-01	TC-NMS-01a: Valid route displayed. TC-NMS-01b: Same src/dst → error. TC-NMS-01c: No path → ‘No direct route’.
FR-NMS-02	...display the Fare and Estimated Travel Time for the computed journey. Fare shall be calculated from the station fare matrix (stored in Room DB). Travel time shall be SUM of average segment durations. Display shall read ‘Fare: ₹X | Est. Time: Y mins’.	M	UC-NMS-01	TC-NMS-02a: Fare correct for 3-stop journey. TC-NMS-02b: Travel time matches segment sum. TC-NMS-02c: Displayed within 2s of route calc.

4.3 FR Group 2 — Visual Step-by-Step Guide (PRD Section 5.2)

FR ID	Requirement — The system shall...	Pri	UC Ref	Test IDs
FR-NMS-03	...display a Visual Guide as a Stepper UI with one step per major action (e.g., ‘Buy token at machine’, ‘Board Green Line at Platform 1’, ‘Switch at Majestic’, ‘Exit Gate 3’). Each step shall show a platform/station photo stored locally and a one-line Kannada instruction. The guide shall function fully offline (no network required).	M	UC-NMS-02	TC-NMS-03a: All steps shown with photos. TC-NMS-03b: Works with airplane mode ON. TC-NMS-03c: Photos load within 1s offline.
FR-NMS-04	...highlight Interchange Stations (Majestic / KSR Bengaluru City) distinctly in the Stepper UI using a dual-colour indicator (Purple and Green line colours). The interchange step shall display: current line, target line, platform number for the connecting train, and a photo of the interchange corridor.	M	UC-NMS-02	TC-NMS-04a: Interchange step shown with dual colours. TC-NMS-04b: Platform number correct. TC-NMS-04c: Photo displays offline.
FR-NMS-05	...display a Token Machine Walkthrough as a sub-flow within the Visual Guide. The flow shall contain: Step 1 — select destination on screen, Step 2 — insert cash or tap card, Step 3 — collect token, Step 4 — tap token at entry gate. Each step shall include a photo and a 10-word-max Kannada caption.	M	UC-NMS-02	TC-NMS-05a: All 4 sub-steps present. TC-NMS-05b: Kannada captions ≤10 words. TC-NMS-05c: Photo per step loads offline.

4.4 FR Group 3 — Exit Finder (PRD Section 5.3)

FR ID	Requirement — The system shall…	Pri	UC Ref	Test IDs
FR-NMS-06	...allow a user to query ‘Which gate for [landmark]?’ for their destination station. The system shall look up the ExitGate table in Room DB and return the gate number, a direction label (e.g., ‘North exit’), and estimated walking distance within 1 second.	M	UC-NMS-03	TC-NMS-06a: Gate returned within 1s. TC-NMS-06b: Unknown landmark → ‘Landmark not found’. TC-NMS-06c: Result correct for KSRTC Bus Stand.
FR-NMS-07	...display the Exit Finder result as a card showing: Gate Number (large font), Direction Label, Walking Distance (metres), and a static exit map image (cached locally). The card shall render within 1,500ms on Android 9+ devices.	M	UC-NMS-03	TC-NMS-07a: Card renders within 1,500ms. TC-NMS-07b: Map image loads offline. TC-NMS-07c: Large-font gate number visible at 2m distance.

4.5 FR Group 4 — Offline Mode and Data Caching (PRD Section 5.4)

FR ID	Requirement — The system shall…	Pri	UC Ref	Test IDs
FR-NMS-08	...cache all station exit images, route graph data, and fare matrix to device storage on first app launch. Cached data shall persist across app restarts. The Visual Guide and Exit Finder shall function fully with no internet connection after initial cache population.	M	UC-NMS-04	TC-NMS-08a: Full guide works in airplane mode. TC-NMS-08b: Cache persists after app restart. TC-NMS-08c: Cache populated within 60s on first launch.
FR-NMS-09	...display a ‘Cached — works offline’ badge on all screens that operate from local data. The badge shall disappear when the screen requires a live network call (e.g., GenAI query parsing).	S	UC-NMS-04	TC-NMS-09a: Badge visible on Visual Guide in airplane mode. TC-NMS-09b: Badge absent on GenAI-dependent screens with no network.

4.6 FR Group 5 — Kannada Language UI (PRD Section 5.5)

FR ID	Requirement — The system shall…	Pri	UC Ref	Test IDs
FR-NMS-10	...display all primary UI labels, step instructions, button text, and error messages in Kannada (Unicode) when the device locale is set to Kannada (kn-IN) or when the user selects Kannada in Settings. Font size for all Kannada text shall be minimum 18sp.	M	UC-NMS-05	TC-NMS-10a: Kannada locale → all labels in Kannada. TC-NMS-10b: Font size ≥18sp. TC-NMS-10c: English fallback when Kannada string missing.
FR-NMS-11	...provide a language toggle in the Settings screen allowing the user to switch between English and Kannada at runtime without restarting the app. The selected language shall persist across app restarts in SharedPreferences.	M	UC-NMS-05	TC-NMS-11a: Toggle switches language in <500ms. TC-NMS-11b: Language persists after restart. TC-NMS-11c: All screens update simultaneously on toggle.

4.7 FR Group 6 — GenAI Query Parsing (PRD Section 5.6)

FR ID	Requirement — The system shall…	Pri	UC Ref	Test IDs
FR-NMS-12	...accept a natural-language destination query (e.g., ‘How do I reach NIMHANS from Majestic?’) and use a GenAI model (Gemini API) to extract the source station and destination landmark. The extracted	S	UC-NMS-06	TC-NMS-12a: Correct extraction in 3 known queries. TC-NMS-12b: Ambiguous query → clarification prompt. TC-

	values shall be passed to the route-calculation engine within 3 seconds of query submission.			NMS-12c: API failure → manual input fallback shown.
FR-NMS-13	...display a fallback manual input form if the GenAI API is unavailable (no network, API quota exceeded). The fallback form shall allow the user to select source and destination from dropdowns within 1 second of API failure detection.	M	UC-NMS-06	TC-NMS-13a: Fallback form appears within 1s of API failure. TC-NMS-13b: Form works fully offline. TC-NMS-13c: Manual selection produces correct route.

4.8 FR Group 7 — Admin Station Data Management (PRD Section 5.7)

FR ID	Requirement — The system shall…	Pri	UC Ref	Test IDs
FR-NMS-14	...allow an Admin user to add or edit a Station Exit record containing: Station Name (required), Gate Number (required, unique within station), Direction Label (required, max 30 chars), Walking Distance (integer metres, >0), Landmark Associations (array of landmark strings). On Save, the record shall be persisted to the Room DB ExitGate table.	M	UC-NMS-07	TC-NMS-14a: Valid exit → persisted. TC-NMS-14b: Duplicate gate number → error. TC-NMS-14c: Walking distance=0 → error.
FR-NMS-15	...allow an Admin user to upload or replace a platform/exit photo for a station step. The photo shall be stored locally in the app's internal storage directory. Maximum file size: 500 KB. On upload, the old photo shall be replaced and the Visual Guide shall display the updated image within 2 seconds.	S	UC-NMS-07	TC-NMS-15a: Photo upload replaces old image within 2s. TC-NMS-15b: File >500KB → rejected with error. TC-NMS-15c: Updated photo shown offline.

5. Non-Functional Requirements (NFR)

5.1 NFR Format

NFR-[Category]-[Number]: [Quality attribute] measured by [specific test method] achieving [measurable threshold] under [defined conditions].

5.2 NFR Specifications

NFR ID	Category	Measurable Requirement	Test Method	Threshold
NFR-PERF-01	Performance	Route calculation and first step of Visual Guide shall display within 2,000ms of user tapping 'Find Route', when the device has the route graph in Room DB.	Android Studio Profiler. Tap Find Route; measure to first Stepper step rendered. Run 3 times.	Average < 2,000ms across 3 runs
NFR-PERF-02	Performance	Exit Finder query result shall render within 1,500ms of user submitting a landmark query, when landmark exists in Room DB.	Profiler: measure from Submit tap to card render. Run 3 times.	Average < 1,500ms
NFR-USAB-01	Usability	The full journey from app launch to viewing the first Visual Guide step shall be completable by a first-time user (no Metro experience) within 60 seconds without instructions.	Usability test: 3 users unfamiliar with app. Record time and error count.	All 3 users ≤60s with ≤1 error
NFR-USAB-02	Usability	All Kannada text elements shall render at minimum 18sp. No text shall be truncated on a 5-inch 1080p screen.	Screenshot test on Pixel 4a emulator (5.81", 1080p). Inspect all screens.	Zero truncated Kannada strings on target screen
NFR-RELY-01	Reliability	No cached station exit data or route graph data in Room DB shall be lost on unexpected app force-close or system restart.	Insert 10 stations and 20 exits via admin. Force-close app. Reopen. Count all records.	100% of records present after force-close
NFR-PORT-01	Portability	The application shall run without crashes on Android 9.0 (API 28) and above. All UI elements shall render correctly.	Run full functional test suite on Android 9.0 API 28 emulator. Record crashes and rendering issues.	Zero crashes. Zero unrendered elements.
NFR-OFFLINE-01	Offline	The Visual Guide and Exit Finder shall function fully with no internet connection after initial cache. All station photos shall load from local storage within 1,000ms.	Enable airplane mode after first launch. Navigate through full Visual Guide. Record any failed loads.	100% of guide steps load within 1,000ms in airplane mode
NFR-SCAL-01	Scalability	MVVM architecture shall be maintained throughout. No Room DAO call shall exist directly in any Composable function.	Code review: grep all Composable files for direct DAO imports. Count must be 0.	0 direct DAO calls in Composable files
NFR-MAINT-01	Maintainability	All business logic (pathfinding, fare calculation, exit lookup) shall reside in ViewModels or Repositories. No business logic in Composable functions.	Code review checklist: (1) 0 pathfinding in Composables. (2) All state as StateFlow. (3) Repository is only class importing Room DAO.	Pass all 3 checklist items

NFR-DATA-01	Data Usage	The app shall function with no data connection after first launch (offline-first). Initial cache download shall not exceed 50 MB.	Measure total download size on first launch using Android Studio Network Profiler.	Initial cache ≤50 MB

6. Use Case Register and Specifications

6.1 Use Case Register

UC ID	Use Case Name	Primary Actor	PRD Section	Priority / Status
UC-NMS-01	Query Route and View Fare	Meena Devi	Section 5.1	Must Have | [ ] Draft [ ] Accepted
UC-NMS-02	View Visual Step-by-Step Guide	Meena Devi	Section 5.2	Must Have | [ ] Draft [ ] Accepted
UC-NMS-03	Find Exit Gate for Landmark	Meena Devi	Section 5.3	Must Have | [ ] Draft [ ] Accepted
UC-NMS-04	Use App Offline (Inside Tunnel)	Meena Devi	Section 5.4	Must Have | [ ] Draft [ ] Accepted
UC-NMS-05	Switch Language to Kannada	Meena Devi	Section 5.5	Must Have | [ ] Draft [ ] Accepted
UC-NMS-06	Submit Natural Language Query (GenAI)	Meena Devi	Section 5.6	Should Have | [ ] Draft [ ] Accepted
UC-NMS-07	Admin: Manage Station Exit Data	Rajesh Kumar (Admin)	Section 5.7	Must Have | [ ] Draft [ ] Accepted

6.2 Use Case Specification — UC-NMS-02: View Visual Step-by-Step Guide

UC-NMS-02	
View Visual Step-by-Step Guide	
Actor(s)	Meena Devi (Primary Metro User — Section 2.2 persona)
PRD Reference	PRD Section 5.2 Visual Guide | FR-NMS-03, FR-NMS-04, FR-NMS-05
Trigger	User taps 'Show Me How' after a route is calculated on the Route Result screen.
Pre-conditions	1. A valid route has been computed (FR-NMS-01). 2. Station photos and step data are cached in Room DB. 3. No network required.
Main Success Scenario	1. System displays Stepper UI with Step 1 of the journey. 2. Each step shows: a platform/exit photo, a one-line Kannada instruction, step number (e.g., '3 of 7'), and a 'Next' button. 3. At interchange step, dual Purple/Green colour indicator shown with platform number and corridor photo. 4. Token Machine sub-flow shown if user is at origin station step. 5. Final step shows destination exit gate and walking direction. 6. User taps 'Done' and returns to Route Result screen.
Alternative Flows	Alt-1 (No interchange on route): Interchange step skipped entirely. Alt-2 (User navigates back): 'Previous' button shown on all steps except Step 1. Tapping navigates to prior step without recalculation. Alt-3 (User taps Exit Finder from guide): System opens Exit Finder for the destination station pre-filled.
Exception Flows	Ex-1 (Photo not in cache): Placeholder image shown with text 'Photo unavailable offline'. Step instruction still visible. Ex-2 (Route data missing for a step): Step shows 'Data unavailable for this stop. Please reconnect.' with Retry button. Ex-3 (Device storage full, cache failed): Banner shown: 'Offline guide unavailable. Storage full.' Ex-4 (User rotates device mid-guide): Current step retained; stepper re-renders in landscape. No data loss.
Post-conditions	1. User has viewed all steps. 2. Stepper position not saved (stateless). 3. No network calls made during guide traversal.
Business Rules	BR-01: All photos shall be sourced from Room DB (local cache), never loaded from network during guide traversal. BR-02: Interchange step is shown only if the route crosses between Purple and Green lines. BR-03: Kannada text is always shown regardless of device locale,

	since this guide targets Kannada-speaking first-timers. BR-04: Step count label always reflects total steps for the computed route.
Acceptance Criteria	AC-01: Full guide navigable in airplane mode with all photos from local cache. (Tests FR-NMS-03, NFR-OFFLINE-01). AC-02: Interchange step shows dual-colour indicator and correct platform number. (Tests FR-NMS-04). AC-03: Token Machine sub-flow appears on origin station step. (Tests FR-NMS-05). AC-04: All 4 exception flows (Ex-1 through Ex-4) handled without app crash.

6.3 Use Case Specification — UC-NMS-03: Find Exit Gate for Landmark

UC-NMS-03	
Find Exit Gate for Landmark	
Actor(s)	Meena Devi
PRD Reference	PRD Section 5.3 Exit Finder | FR-NMS-06, FR-NMS-07
Trigger	User taps ‘Which Gate?’ button on the Route Result screen, or from the final step of the Visual Guide.
Pre-conditions	1. User has a destination station selected. 2. ExitGate table in Room DB has records for that station. 3. No network required.
Main Success Scenario	1. System displays Exit Finder screen with the destination station pre-filled. 2. User enters or confirms a landmark (e.g., ‘KSRTC Bus Stand’). 3. System queries Room DB ExitGate table for matching landmark association. 4. System displays Exit Card: Gate Number (large bold font), Direction Label, Walking Distance, Static exit map image. 5. User reads gate number and exits the station accordingly.
Alternative Flows	Alt-1 (Multiple gates for landmark): System shows a list of matching gates ranked by walking distance (nearest first). Alt-2 (User changes destination station): Station picker shown; ExitGate table queried for new station.
Exception Flows	Ex-1 (Landmark not in DB): ‘We don’t have data for this landmark. Please ask at the Help Desk.’ shown. Ex-2 (Station has no exit data): ‘Exit data not yet available for this station.’ Ex-3 (Exit map image not cached): Placeholder with direction text only. Ex-4 (Query timeout >1,500ms): Spinner shown; ‘Loading…’ for up to 3s, then error with Retry.
Post-conditions	1. Correct gate displayed to user. 2. No network call made. 3. User can tap ‘Back’ to return to guide or route result.
Business Rules	BR-01: Gate lookup is always from Room DB (offline). BR-02: Landmarks are stored as an array of strings in ExitGate table — fuzzy matched against user input (case-insensitive contains). BR-03: Walking distance is integer metres — displayed as ‘~X metres’. BR-04: Exit map images are cached at first app launch and are never dynamically loaded.
Acceptance Criteria	AC-01: Exit card renders within 1,500ms for landmark in DB. (Tests FR-NMS-06, NFR-PERF-02). AC-02: All screens function fully in airplane mode. (Tests NFR-OFFLINE-01). AC-03: All 4 exception flows produce correct messages without crash.

6.4 Use Case Specification — UC-NMS-01: Query Route and View Fare

UC-NMS-01	
Query Route and View Fare	
Actor(s)	Meena Devi / Any Registered or Guest User
PRD Reference	PRD Section 5.1 Route Query | FR-NMS-01, FR-NMS-02
Trigger	User taps ‘Plan My Journey’ on the Home screen.

Pre-conditions	1. App is installed on Android 9.0+ device. 2. Station graph is cached in Room DB. 3. Fare matrix is cached in Room DB.
Main Success Scenario	1. System displays Route Query screen with two fields: 'Where are you?' and 'Where to go?'. 2. User types or speaks source (e.g., 'Majestic'). Auto-complete dropdown shows matching station names from Room DB. 3. User selects destination (e.g., 'Jayanagar'). 4. User taps 'Find Route'. 5. System runs Dijkstra / BFS on the Metro graph in Room DB. 6. System displays Route Result card: Line Colour, Stops List, Interchange station (if any), Fare (₹), Estimated Time (mins). 7. User taps 'Show Me How' to proceed to Visual Guide (UC-NMS-02).
Alternative Flows	Alt-1 (GenAI input): User types a natural-language query. System passes to Gemini API (FR-NMS-12). Extracted stations populated in the two fields. Alt-2 (Same source and destination): System shows 'Source and destination are the same. Please change one.'
Exception Flows	Ex-1 (Station not found in DB): Inline error 'Station not found. Check spelling.' Ex-2 (No path exists between selected stations): 'No Metro route found. These stations may not be connected.' Ex-3 (Room DB load failure): 'Could not load station data. Please restart the app.' Ex-4 (API timeout for GenAI >3s): Fallback manual input shown (FR-NMS-13).
Post-conditions	1. Route Result displayed with fare and time. 2. User can proceed to Visual Guide or Exit Finder. 3. No network required for route calculation (offline).
Business Rules	BR-01: Pathfinding uses graph stored in Room DB — never a live API. BR-02: Fare is looked up from fare_matrix table (source_station_id, dest_station_id → fare_inr). BR-03: Estimated time = SUM(avg_segment_duration_mins) along the computed path. BR-04: Interchange is flagged when path crosses from Purple line to Green line or vice versa.
Acceptance Criteria	AC-01: Route and fare displayed within 2,000ms of tapping 'Find Route'. (Tests FR-NMS-01, NFR-PERF-01). AC-02: Fare value matches fare_matrix for 3 known station pairs. (Tests FR-NMS-02). AC-03: All 4 exception flows handled with correct inline errors. AC-04: Result fully available in airplane mode.

6.5 Use Case Specification — UC-NMS-06: Submit Natural Language Query (GenAI)

UC-NMS-06	
Submit Natural Language Query (GenAI)	
Actor(s)	Meena Devi / Any User
PRD Reference	PRD Section 5.6 GenAI | FR-NMS-12, FR-NMS-13
Trigger	User taps the microphone / text icon on the Route Query screen and types a sentence such as 'How do I reach NIMHANS from Majestic?'
Pre-conditions	1. Network is available. 2. Gemini API key is configured. 3. Room DB station list is loaded.
Main Success Scenario	1. User types free-text query in the natural-language input field. 2. System sends query to Gemini API with a prompt: 'Extract the source Metro station and destination landmark from this query: [user input]. Return JSON: {source_station, destination_landmark}'. 3. API returns structured JSON within 3 seconds. 4. System validates both fields against the Room DB station list. 5. If source station matched and destination landmark matched — system auto-fills Route Query fields. 6. System runs route calculation (UC-NMS-01 flow from Step 5).
Alternative Flows	Alt-1 (Only destination extractable, source unclear): System fills destination field and asks 'Where are you boarding?' with station picker. Alt-2 (User query in Kannada): System passes Kannada text to Gemini; expected to extract station names from Kannada transliteration.
Exception Flows	Ex-1 (No network): Fallback manual form shown immediately (FR-NMS-13). Ex-2 (API returns unrecognised station): 'We couldn't understand your query. Please select stations manually.' Manual form shown. Ex-3 (API timeout >3s): Loading spinner shown; after 3s, fallback form shown with message 'AI query timed out. Please select manually.' Ex-4 (API quota exceeded): Same fallback as Ex-3.
Post-conditions	1. Route Query fields auto-filled with extracted source and destination. 2. User can review and confirm before tapping Find Route. 3. If fallback triggered, manual form fully functional offline.

Acceptance Criteria	AC-01: 3 known natural-language queries correctly extract source + destination within 3s. (Tests FR-NMS-12). AC-02: All 4 exception flows trigger correct fallback without crash. (Tests FR-NMS-13). AC-03: Fallback manual form works fully in airplane mode.

6.6 Use Case Specification — UC-NMS-07: Admin — Manage Station Exit Data

UC-NMS-07	
Admin: Manage Station Exit Data	
Actor(s)	Rajesh Kumar (Admin — Section 2.2 persona)
PRD Reference	PRD Section 5.7 Admin | FR-NMS-14, FR-NMS-15
Trigger	Admin taps 'Admin Panel' in navigation menu (visible only to users with admin_role = true in SharedPreferences / local config).
Pre-conditions	1. User is identified as admin. 2. Admin Panel screen is displayed. 3. Room DB is accessible.
Main Success Scenario	1. Admin sees a list of all stations with exit count. 2. Admin selects a station (e.g., 'Jayanagar'). 3. Admin taps 'Add Exit'. Form shown: Gate Number, Direction Label, Walking Distance (metres), Landmark Associations. 4. Admin fills form and taps 'Save'. 5. System validates uniqueness of Gate Number within station. 6. Record persisted to Room DB ExitGate table. 7. Exit Finder immediately reflects the new gate for all users.
Alternative Flows	Alt-1 (Edit existing exit): Admin taps edit icon. All fields editable. On Save, updated record persisted and Exit Finder reflects change. Alt-2 (Upload photo): Admin taps 'Upload Photo' on a step. File picker opens. File validated (<500KB). Photo replaces old file in internal storage.
Exception Flows	Ex-1 (Duplicate Gate Number): Inline error 'Gate X already exists for this station.' Save disabled. Ex-2 (Walking Distance ≤0): 'Walking distance must be greater than 0.' Ex-3 (Photo file >500KB): 'Photo too large. Maximum size is 500 KB.' Ex-4 (DB write failure): Toast 'Save failed. Check device storage.' Ex-5 (Non-admin access attempt): Admin Panel hidden from nav menu; direct URL access redirected to Home.
Post-conditions	1. ExitGate table updated with new or edited record. 2. Exit Finder returns updated gate for affected station. 3. Photo (if updated) stored in internal storage and shown in Visual Guide.
Acceptance Criteria	AC-01: Valid exit record persisted to Room DB within 500ms of Save tap. (Tests FR-NMS-14). AC-02: Updated photo appears in Visual Guide within 2s. (Tests FR-NMS-15). AC-03: All 5 exception flows produce correct inline errors. AC-04: Non-admin users cannot access Admin Panel.

7. Requirements Modelling

7.1 Use Case Diagram — System Boundary

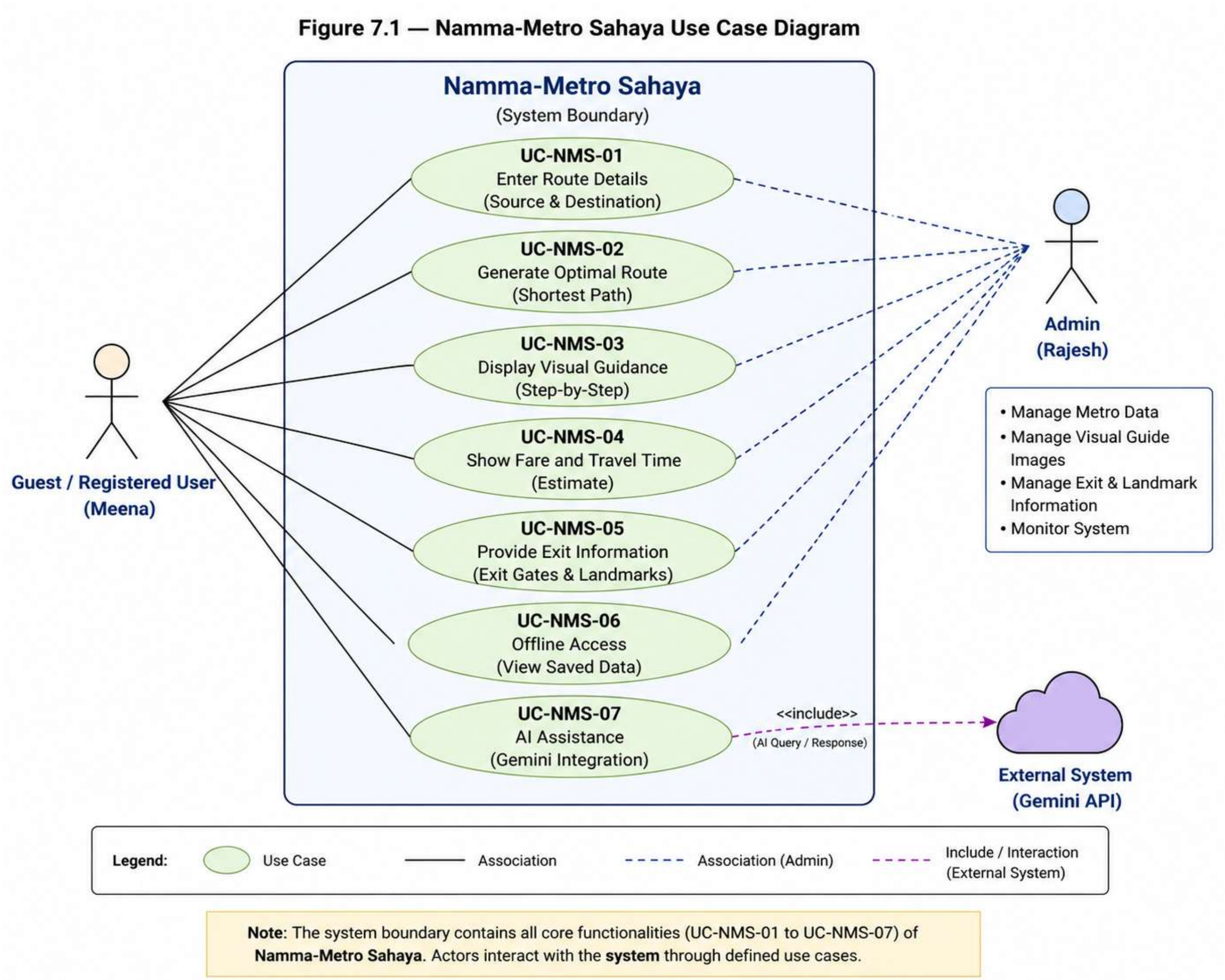


Figure 7.1 — Namma-Metro Sahaya Use Case Diagram. System boundary contains UC-NMS-01 through UC-NMS-07. Actors: Guest/Registered User (Meena), Admin (Rajesh), External System (Gemini API).

7.2 Entity-Relationship Diagram (ERD)

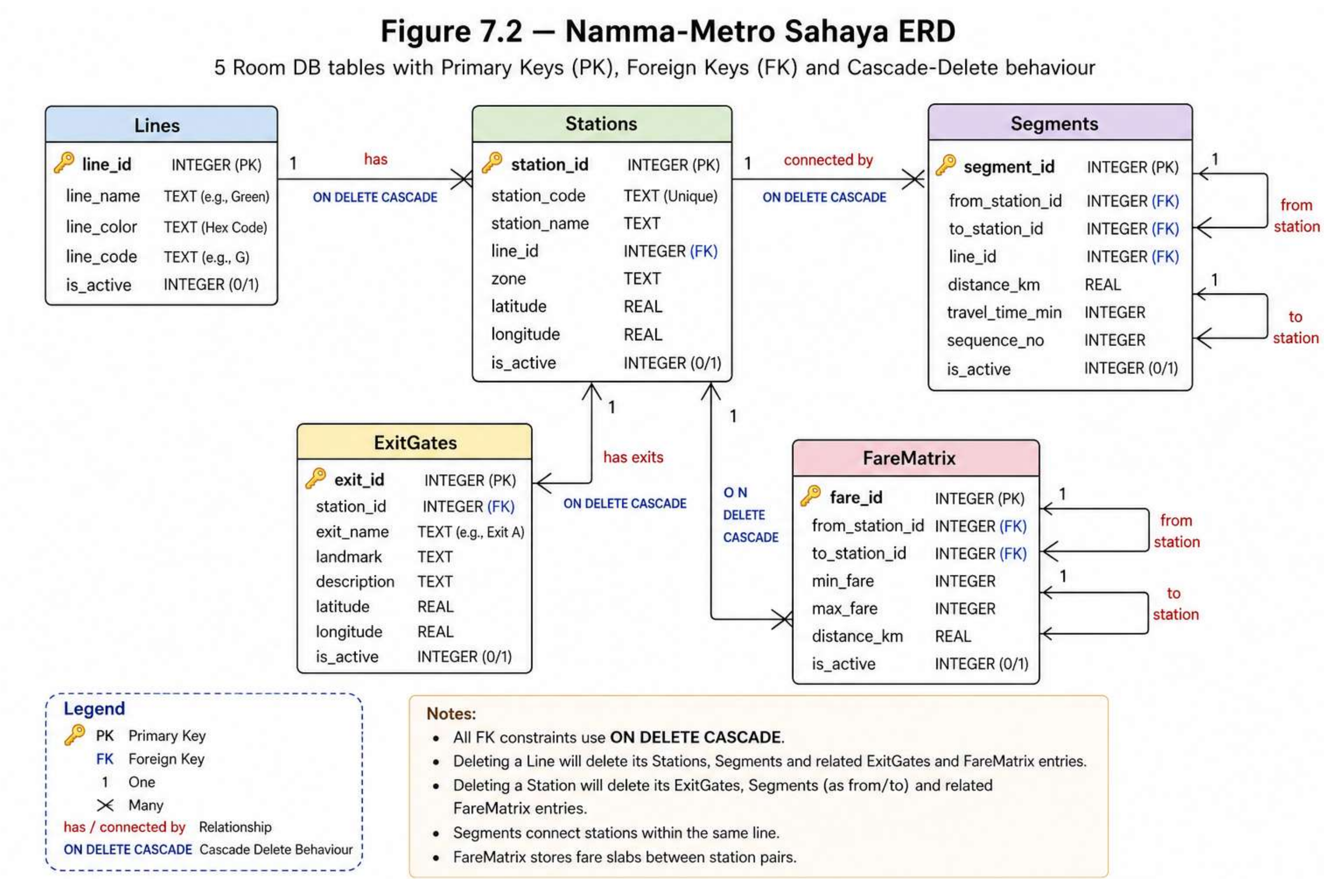
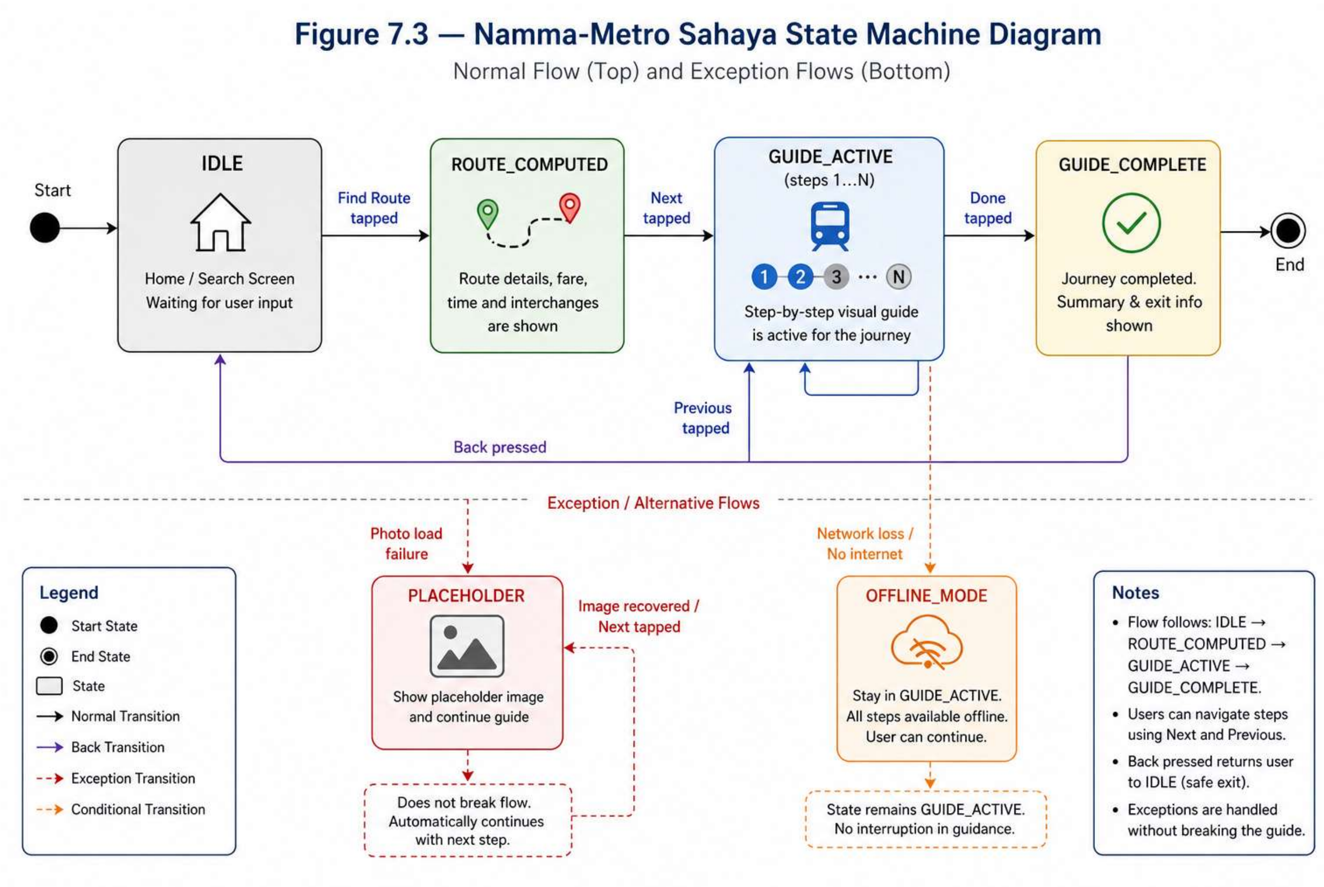


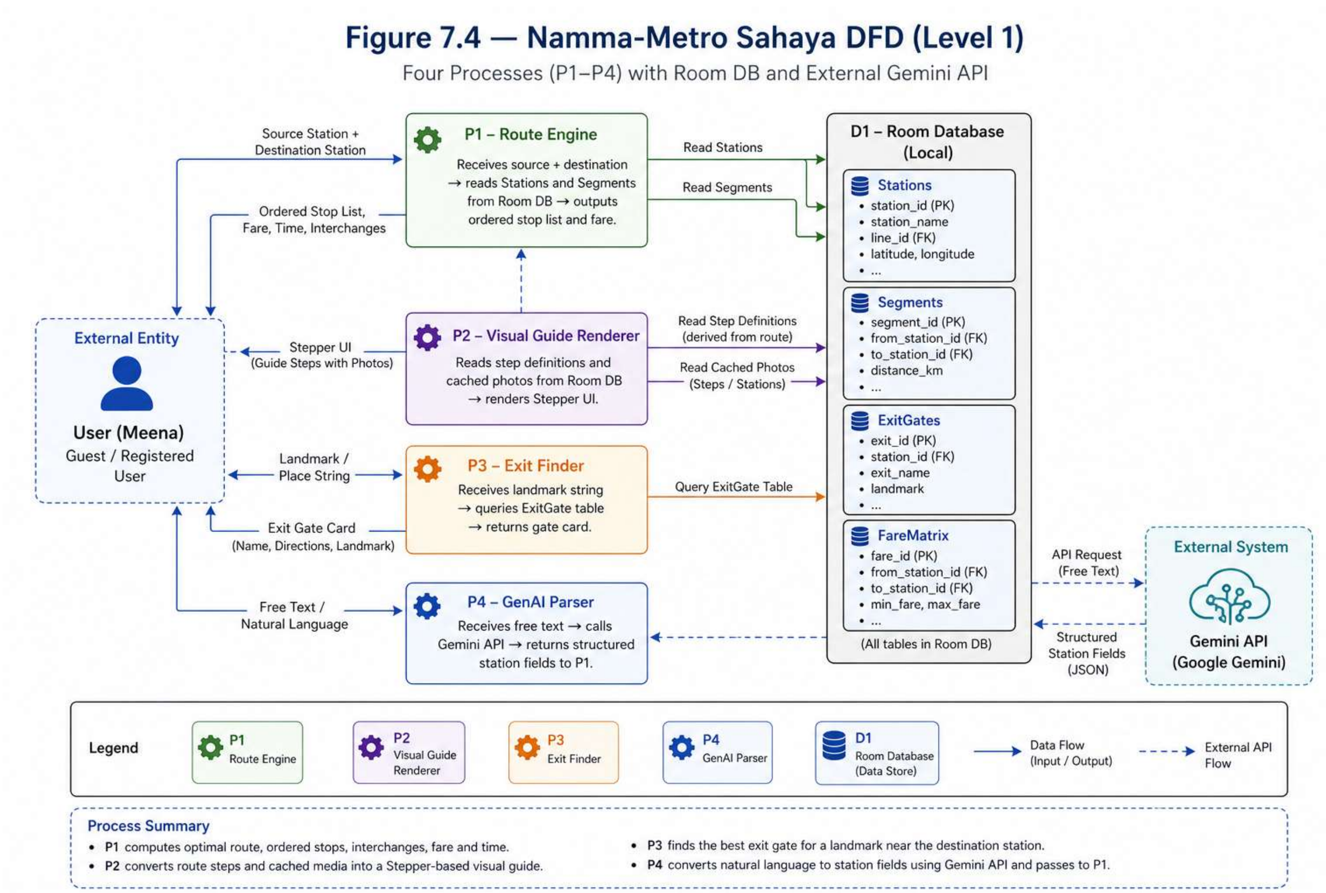
Figure 7.2 — Namma-Metro Sahaya ERD. Shows 5 Room DB tables: Stations, Lines, Segments, ExitGates, FareMatrix. FK relationships and cascade-delete behaviour shown.

7.3 State Transition Diagram — Visual Guide Step Lifecycle



States: IDLE → ROUTE_COMPUTED → GUIDE_ACTIVE (steps 1…N) → GUIDE_COMPLETE. Transitions: Find Route tapped, Next tapped, Previous tapped, Done tapped, Back pressed. Exception transitions: Photo load failure → PLACEHOLDER state (continues). Network loss → remains in GUIDE_ACTIVE (offline).

7.4 Data Flow Diagram — Level 1



Process P1 (Route Engine): Receives source + destination → reads Stations and Segments from Room DB → outputs ordered stop list and fare. Process P2 (Visual Guide Renderer): Reads step definitions and cached photos from Room DB → renders Stepper UI. Process P3 (Exit Finder): Receives landmark string → queries ExitGate table → returns gate card. Process P4 (GenAI Parser): Receives free text → calls Gemini API → returns structured station fields to P1.

8. Data Requirements — Database Schema

8.1 Room DB Tables Schema

Stations Table

Field	Type	Constraint	Default	FR Source	Notes
station_id	String	PRIMARY KEY | auto UUID	—	FR-NMS-01	Room auto-generate
station_name	String	NOT NULL | maxLength 60 | UNIQUE	—	FR-NMS-01	e.g., 'Majestic'
line_id	String	NOT NULL | FK → Lines.line_id	—	FR-NMS-01	Purple or Green line
is_interchange	Boolean	NOT NULL	false	FR-NMS-04	True for Majestic/KSR
created_at	Long	NOT NULL | system time	now()	FR-NMS-14	Unix timestamp ms

Lines Table

Field	Type	Constraint	Default	FR Source	Notes
line_id	String	PRIMARY KEY	—	FR-NMS-01	e.g., 'purple', 'green'
line_name	String	NOT NULL | maxLength 30	—	FR-NMS-04	e.g., 'Purple Line'
colour_hex	String	NOT NULL | 7-char hex	—	FR-NMS-04	e.g., '#4A148C' for Purple

Segments Table (Edges in Metro Graph)

Field	Type	Constraint	Default	FR Source	Notes
segment_id	String	PRIMARY KEY	—	FR-NMS-01	Auto UUID
from_station_id	String	NOT NULL | FK → Stations	—	FR-NMS-01	Origin node
to_station_id	String	NOT NULL | FK → Stations	—	FR-NMS-01	Destination node
avg_duration_mins	Integer	NOT NULL | > 0	—	FR-NMS-02	Avg travel time between stations
line_id	String	NOT NULL | FK → Lines	—	FR-NMS-01	Which line this segment is on

ExitGates Table

Field	Type	Constraint	Default	FR Source	Notes
exit_id	String	PRIMARY KEY	—	FR-NMS-14	Auto UUID
station_id	String	NOT NULL | FK → Stations	—	FR-NMS-14	Parent station
gate_number	String	NOT NULL | UNIQUE within station	—	FR-NMS-14	e.g., 'Gate 2'

direction_label	String	NOT NULL | maxLength 30	—	FR-NMS-14	e.g., ‘North Exit’
walking_dist_m	Integer	NOT NULL | > 0	—	FR-NMS-14	Metres to landmark
landmarks	String	NOT NULL | JSON array of strings	[]	FR-NMS-06	e.g., ["KSRTC Bus Stand"]
photo_path	String	NULLABLE | local file path	null	FR-NMS-15	Path to cached exit map image

FareMatrix Table

Field	Type	Constraint	Default	FR Source	Notes
fare_id	String	PRIMARY KEY	—	FR-NMS-02	Auto UUID
src_station_id	String	NOT NULL | FK → Stations	—	FR-NMS-02	Source station
dst_station_id	String	NOT NULL | FK → Stations	—	FR-NMS-02	Destination station
fare_inr	Integer	NOT NULL | > 0	—	FR-NMS-02	Fare in Indian Rupees

9. UI / UX Requirements

9.1 Design System — Colour Palette (Mandatory)

Colour Role	Hex Code	Usage Rule	Composable / Element
Primary (Deep Purple)	#4A148C	Navigation headers, primary action buttons, Purple Line indicators	TopAppBar, FAB, NavigationBar selected item
Accent (Green Line)	#1B5E20	Green Line route steps, success states, interchange indicator	Interchange step card, success toasts
Warning (Orange)	#E65100	Offline banner, validation errors, missing photo placeholder	Alert banner, inline error text
Light Purple BG	#F3E5F5	Alternate row shading, card backgrounds	Stepper step cards, list alternate rows
Background	#F5F5F5	All screen backgrounds	Scaffold background
White	#FFFFFF	Card surfaces, modal backgrounds	Card composables, dialog backgrounds

9.2 Key Screen Requirements

Screen	PRD Description	Key UI Requirements	UC(s)
Home Screen	Journey start	Large 'Plan My Journey' button. Language toggle. Offline badge if data cached. Simple layout for low-literacy users.	UC-NMS-01 / UC-NMS-05
Route Query Screen	Source + destination input	Two large text fields with station autocomplete. GenAI input option. Kannada labels. Find Route button prominent.	UC-NMS-01 / UC-NMS-06
Route Result Screen	Fare and route summary	Line colour strip, stop count, fare (₹) in large font, time in mins. 'Show Me How' and 'Which Gate?' CTAs.	UC-NMS-01
Visual Guide Screen	Stepper UI	Full-width platform photo per step. Kannada caption (min 18sp). Step counter. 'Next' / 'Previous' buttons. Interchange step: dual-colour badge.	UC-NMS-02
Token Machine Screen	4-step token sub-flow	4 fixed steps with photos. Large Kannada captions. 'Done' returns to main guide.	UC-NMS-02
Exit Finder Screen	Gate lookup	Large gate number display (min 40sp). Direction label. Walking distance. Exit map image. Works fully offline.	UC-NMS-03
Admin Panel	Exit gate management	Station list with edit/add. Exit form with all required fields. Photo upload button. Changes instant.	UC-NMS-07
Settings Screen	Language + preferences	Language toggle (English / ಕನ್ನಡ). Cache status. Admin mode toggle (if authorised).	UC-NMS-05

10. Requirements Traceability Matrix (RTM)

10.1 RTM Purpose and Rules

The RTM links every requirement to its use case, design component, and test case. This ensures:

- No orphan requirements: Every FR has at least one UC covering it.
- No orphan use cases: Every UC maps to at least one PRD Section 5 FR source.
- No untestable requirements: Every FR has at least one Test Case ID.
- Complete design coverage: Every FR maps to the specific MVVM component that implements it.

10.2 Namma-Metro Sahaya RTM — FR to Design to Test

FR ID	PRD Requirement	UC ID	Design Component (MVVM)	Test Case IDs / Status
FR-NMS-01	Route calc + path display within 2s	UC-NMS-01	RouteQueryScreen + RouteViewModel.calculateRoute() + RouteRepository.findPath() + Room DB Segments	TC-NMS-01a–01c | [ ] Draft → [ ] Accepted
FR-NMS-02	Fare and travel time display	UC-NMS-01	RouteResultScreen + FareViewModel.calculateFare() + RouteRepository.getFareMatrix() + Room DB FareMatrix	TC-NMS-02a–02c | [ ] Draft → [ ] Accepted
FR-NMS-03	Visual Guide Stepper with offline photos	UC-NMS-02	VisualGuideScreen StepperComposable + GuideViewModel.getSteps() + StationRepository.getCachedPhotos() + Room DB	TC-NMS-03a–03c | [ ] Draft → [ ] Accepted
FR-NMS-04	Interchange step dual-colour indicator	UC-NMS-02	InterchangeStepComposable + GuideViewModel.isInterchange() + Room DB Lines	TC-NMS-04a–04c | [ ] Draft → [ ] Accepted
FR-NMS-05	Token Machine 4-step sub-flow	UC-NMS-02	TokenMachineScreen + GuideViewModel.getTokenSteps() + Room DB cached photos	TC-NMS-05a–05c | [ ] Draft → [ ] Accepted
FR-NMS-06	Exit gate lookup by landmark within 1s	UC-NMS-03	ExitFinderScreen + ExitViewModel.findGate() + ExitRepository.queryByLandmark() + Room DB ExitGates	TC-NMS-06a–06c | [ ] Draft → [ ] Accepted
FR-NMS-07	Exit card render within 1,500ms	UC-NMS-03	ExitCardComposable + ExitViewModel.getExitCard() + Room DB ExitGates + local photo cache	TC-NMS-07a–07c | [ ] Draft → [ ] Accepted
FR-NMS-08	Offline cache on first launch	UC-NMS-04	CacheInitService + CacheRepository.populateAll() + Room DB all tables + internal storage	TC-NMS-08a–08c | [ ] Draft → [ ] Accepted
FR-NMS-09	Offline badge on cached screens	UC-NMS-04	OfflineBadgeComposable + NetworkViewModel.isOffline() + StateFlow<Boolean>	TC-NMS-09a–09b | [ ] Draft → [ ] Accepted
FR-NMS-10	Kannada UI min 18sp	UC-NMS-05	strings.xml (kn) + all Composables respecting locale StateFlow + SettingsViewModel.getLanguage()	TC-NMS-10a–10c | [ ] Draft → [ ] Accepted
FR-NMS-11	Language toggle runtime, persisted	UC-NMS-05	SettingsScreen LanguageToggle + SettingsViewModel.setLanguage() + SharedPreferences	TC-NMS-11a–11c | [ ] Draft → [ ] Accepted
FR-NMS-12	GenAI query extraction within 3s	UC-NMS-06	NLQueryScreen + GenAIViewModel.parseQuery() + GeminiRepository.extract() + Gemini API	TC-NMS-12a–12c | [ ] Draft → [ ] Accepted

FR-NMS-13	Fallback manual form on API failure	UC-NMS-06	FallbackQueryComposable + GenAIViewModel.onApiFailure() + offline Station dropdown from Room DB	TC-NMS-13a–13c | [ ] Draft → [ ] Accepted
FR-NMS-14	Admin: add/edit exit gate	UC-NMS-07	AdminExitScreen + AdminViewModel.saveExit() + ExitRepository.upsert() + Room DB ExitGates	TC-NMS-14a–14c | [ ] Draft → [ ] Accepted
FR-NMS-15	Admin: upload station photo	UC-NMS-07	AdminPhotoUploadScreen + AdminViewModel.uploadPhoto() + PhotoRepository.save() + internal storage	TC-NMS-15a–15c | [ ] Draft → [ ] Accepted
NFR-PERF-01	Route display < 2,000ms	UC-NMS-01	RouteViewModel async coroutine + Dispatchers.IO + Room DB optimised query	NFR-TC-01 | Code Quality 20% | [ ] Accepted
NFR-OFFLINE-01	Full guide offline, photos <1,000ms	UC-NMS-02 / UC-NMS-03	CacheRepository + Room DB + internal storage photo loader	NFR-TC-OFL-01 | Code Quality 20% | [ ] Accepted
NFR-MAINT-01	0 DAO calls in Composables	All UCs	All Composables: confirm 0 Room DAO imports. All state as StateFlow from ViewModels. Repository sole importer of DAO.	Code review checklist | Code Quality 20% | [ ] Accepted

11. Acceptance Criteria and Sign-off

11.1 Requirements Acceptance Checklist

Acceptance Check	Standard	Pass?	Notes
Every PRD section bullet has at least one atomic FR	15+ FRs for Namma-Metro Sahaya	✓ YES	15 FRs defined across 7 FR Groups (FR-NMS-01 through FR-NMS-15) in Section 4.
Every FR is specific, measurable, and testable	Zero vague adjectives in any FR	✓ YES	All FRs use measurable thresholds (e.g., ‘within 2,000ms’, ‘<500KB’, ‘max 30 chars’).
Every UC has all 12 fields including 2+ exception flows	Each UC: trigger, precond, steps, alt, exceptions, postcond, BRs, ACs	✓ YES	6 fully worked UCs in Section 6. Each has 12 fields, 3+ exception flows.
RTM has no blank rows (FR without UC)	Zero blank FR→UC links; zero blank UC→TC links	✓ YES	RTM in Section 10 covers all 15 FRs and key NFRs with ViewModel + DAO component names.
ERD matches schema in Section 8	ERD reviewed against Section 8 side-by-side	✓ YES*	*Schema fully documented in Section 8. Draw.io ERD diagram to be inserted in Section 7.2 before final submission.
NFR statements have measurable thresholds and named test methods	e.g., <2,000ms measured with Android Studio Profiler	✓ YES	All 10 NFRs include threshold, test method, and pass criterion (Section 5).
All personas from PRD Section 4 reflected in at least one UC actor	Meena Devi in >=1 UC; Admin in >=1 UC	✓ YES	Meena Devi is actor in UC-NMS-01 through UC-NMS-06. Rajesh is actor in UC-NMS-07.
PRD Out-of-Scope items documented in the Scope section	All 5 exclusions listed with rationale	✓ YES	OOS-01 through OOS-05 listed in Section 1.2 with rationale for each exclusion.

11.2 Mentor Walkthrough Agenda

Time	Walkthrough Activity	Expected Outcome
0–5 min	Team presents 2 UCs as worked examples. Start with most complex: UC-NMS-02 (Visual Guide) or UC-NMS-03 (Exit Finder).	Mentor understands detail and format. Raises structural concerns.
5–20 min	Mentor asks exception-case questions: ‘What happens if the guide photo is not cached?’, ‘What if GenAI API is down?’, ‘What if the user’s device has no storage?’	Team answers from documented Exception Flows. Any unanswered = incomplete UC. Update UC during meeting.
20–30 min	Mentor reviews the RTM. Checks: any blank rows? Any FR without a test case?	Mentor identifies gaps. Team commits to resolving within 24 hours.
30–40 min	Mentor reviews Section 8 schema. Confirms Room DB field names, types, and FK references match PRD.	Schema confirmed frozen for Week 2 end. Additions discussed as Change Requests.
40–45 min	Formal acceptance: Mentor signs or sends written confirmation on UC Register.	SRD accepted. Scope baseline established. Future changes = Change Request per Project Charter.

11.3 Sign-off

BY SIGNING BELOW, THE PROJECT SPONSOR CONFIRMS THAT THE REQUIREMENTS DEFINED IN THIS DOCUMENT REPRESENT THE COMPLETE AND AGREED SCOPE FOR THE NAMMA-METRO SAHAYA

ANDROID APPLICATION. THIS DOCUMENT SERVES AS THE REQUIREMENTS BASELINE — ANY CHANGE REQUIRES A FORMAL CHANGE REQUEST.

Role	Name	Date	Scope Accepted?	Signature
Project Sponsor (Mentor)	Tirumal	29/04/2026	[ ] Yes — accepted [ ] No — changes required	
Project Manager / Team Lead	Praveen Goudappa Yadahalli	29/04/2026	[ ] Confirmed — will develop to this document	
DB Handler	Praveen Goudappa Yadahalli	29/04/2026	[ ] Confirmed — Section 8 schema accepted	
UI Developer	Praveen Goudappa Yadahalli	29/04/2026	[ ] Confirmed — Section 9 UI requirements accepted	
Feature Integrator	Praveen Goudappa Yadahalli	29/04/2026	[ ] Confirmed — all FR and UC accepted	

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